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Low-light image enhancement by deep learning network for improved illumination map

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ABSTRACT

In the case of insufficient illumination conditions, the quality of the captured image is poor. At this time, increasing the brightness of the dark region of the whole image will inevitably aggravate noise pollution. Therefore, the ideal state of low-light image enhancement should be to make the dark areas brighter while not allowing the noise part to be enhanced as well. In this paper, we combine Retinex theory and convolutional neural networks to build a simple and effective model with three sub-modules: the decomposition module, illumination module, and reflection module, in which the decomposition module is used to decompose the input low-light image into illumination and reflection maps, the illumination module improves the feature extraction part of the simple convolutional layer used previously and constitutes a new feature extraction structure by Depthwise separable convolution, which makes up for the shortcoming that some dark areas are not well brightened and better adjusts the brightness of the illumination map, and the reflection module adjusts the texture details more clearly because it adds the illumination map as a reference. In this way, the original space is decoupled into two smaller subspaces to the extent that better results can be achieved. We demonstrate through extensive experiments the effectiveness of the design and its superiority over state-of-the-art techniques, especially in terms of robustness to severe visual defects and flexibility in adjusting luminance.

1. Introduction

In this era of information technology, images have played a very important role in people's daily lives and work, but due to the limitations and influence of the shooting environment (Qiu, 2012; Hsu and Chang, 2010; Provenzi et al., 2014), it often leads to the capture of images with insufficient brightness, lack of clarity, unsaturated colors, and color distortion (see Fig. 1, left side, for more details), so that the effective information obtained from the images is greatly reduced. Usually, the desire to capture high-quality images in dim lighting conditions can be achieved by several operations, such as setting a high ISO, using a long exposure, and turning on the flash, but each has different disadvantages. For example, a high ISO increases the sensitivity of the image sensor to light, but the noise is also amplified, and long exposure is limited to scenes that are shot statically; otherwise, it is likely to lead to a blurred effect. Although image enhancement has made great progress in the past few years, developing an effective and simple low-light image enhancement algorithm that simultaneously increases brightness and suppresses noise remains a challenge.

With the development of machine learning, some low-light image enhancement models based on deep learning have shown good performance. Wei et al. (2018) proposed a deep network model called Retinex-Net, which is the first attempt in the field of low-light image enhancement. By constructing a Retinex theory-based deep learning image decomposition network to perform end-to-end training, the image is decomposed into an illumination map and a reflection map. By adjusting the illumination map the reflection map is directly denoised by the denoising algorithm BM3D (Zhang et al., 2017). Also inspired by the Retinex theory, Wang et al. (2019) proposed a novel progressive Retinex framework that well suppresses the interference caused by an ambiguity between tiny textures and image noise. Zhang et al. (2019) proposed a deep learning network, kindle that decomposes the image into reflection and illumination layers and learns the mapping function, which is more realistic but suffers from color degradation. And based on the kindle network, Zhang et al. (2021a) proposed the KinD+ network, which proposes a new multi-scale illumination attention module(MSIA) that mitigates visual defects (e.g., inhomogeneous speckles and excessive smoothing). Although the KinD+ network fixed the color fusion and degradation that occurred in the kindle network, the simple network structure in the processing module of the illumination map causes KinD+ to still show insufficient brightness and unevenness in the darker regions of the enhanced low-light images, thus missing some details in the dark regions. Deep learning-based methods have shown

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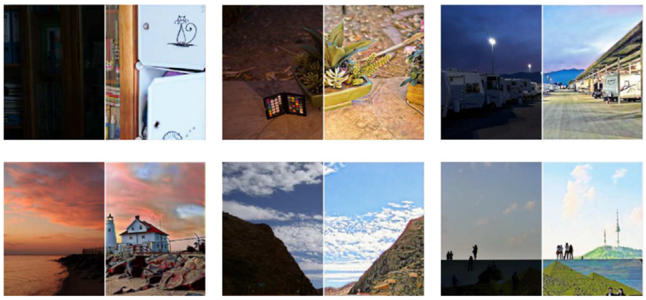

Fig. 1. Comparison of the low-light image (left of the white line) and the enhancement effect of our method (right of the white line). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

superior performance in low-light-enhanced vision tasks, so this study provides a deep learning low-light image enhancement network that solves a practical problem, the main elements of which can be broadly summarized as follows:

- The illumination module improves the feature extraction of simple convolution components used in the past by increasing the network depth of this part, which can better extract the features of low-light images, produce brighter and smoother illumination maps, improve the illumination conditions, and enhance some dark areas in the input low-light images.
- Although increasing network depth enhances the ability to extract image features, it also increases the number of network parameters and the size of the model. So using depthwise separable convolution instead of the previous ordinary convolution and adjusting the features to extract the network structure with reference to the Mobile-Net structure can greatly reduce the size and number of network parameters and achieve the goal of getting a brighter illumination map while minimizing the number of network parameters.
- The reflection module uses a multiscale illumination attention network, which effectively references the improved illumination map and eliminates the visual defect that is magnified by illuminating dark areas. Using a reflection map from the Reflection module, you can make the final enhanced image look more like an image in normal light.

We also conduct more experimental comparisons and comparisons with state-of-the-art algorithms, as shown in Fig. 2, and explore any pertinent circumstances to confirm the superiority of our approach.

2. Related works

2.1. Plain methods

Image enhancement algorithms can be divided into multi-exposure fusion algorithms and single-image contrast enhancement methods based on the number of input multi-exposure images, while the multi-exposure fusion algorithms (Kinoshita and Kiya, 2019; Romero et al., 2015; Fei et al., 2018) required a large number of input multi-exposure samples, which made the computational cost high. In contrast, real images are often obtained once in a specified environment without a specific set of parameters, so we are mainly concerned with single-image contrast enhancement methods. One of the most common early techniques is histogram equalization(HE) (Pizer et al., 1987), which enhanced the contrast of an image by controlling the transformed image histogram so that the pixel values obey a uniform distribution, but the consequences of this may lead to over-enhancement or under-enhancement problems because it is a global process that does not consider the transformation of local luminance. Its variants Dynamic histogram equalization(DHE) (Gu et al., 2015) divided the histogram into several parts and equalized each sub-histogram, and Contrast-limited adaptive histogram equalization(CLAHE) (Yadav, 2014) adaptively limited the degree of contrast enhancement in HE. However, these methods suffer from more severe color dysfunctions, and details in darker regions are often not sufficiently enhanced. Gamma correction(GC) (Hitam et al., 2013) can also be used to adjust the illumination, and although it can be performed nonlinearly at each pixel point individually, this approach does not take into account the relationship between this pixel point and the surrounding neighboring pixels, which leads to a reduced smoothness of the image. Generally speaking, the more advanced implementations are constructed on varied physical priors, among which the Retinex theory (Elad et al., 2003) is the most widely adopted one. Retinex theory argued that there was originally no color in the human world and that color arises due to the interaction between objects and light. Based on this theory, there was also the Single Scale Retinex(SSR) (Liao et al., 2015) method, which constrained the illumination map by Gaussian filtering to make it smooth, and the Multi-Scale Retinex(MSR) (Kornprobst et al., 2016) method used multiscale Gaussian filtering and was an extension of the SSR. LIME (Guo et al., 2017) only processed the illumination map with structured prior knowledge and output the reflection map as the enhanced result. Although these methods can produce good results in some cases, they require manual adjustment of parameters. Also based on Retinex theory, (Wang et al., 2013) proposed a method called NPE that simultaneously enhanced the contrast and maintains the naturalness of the illumination. However, they also usually do not take into account the removal of image noise and sometimes even amplify it. The method proposed by Fu et al. (2016) fused the individual derivatives of the estimated illumination map of the low-light image to adjust the illumination, but this method does not explicitly deal with the color distortion degradation phenomenon, so some texture was lost, which made the whole image less realistic. It also does not take into account the effect of noise on different illumination regions.

Apart from the widely used Retinex theory, many other approaches have also been exploited to accomplish an effective enhancement, such as the method based on dehazing, which exploited the inverse association between the low-light image and the image in a foggy environment to achieve the effect of low-light image enhancement. Although traditional enhancement algorithms have achieved good results in the field of image enhancement, with the rapid development of machine learning, deep learning-based networks have shown superior performance in image enhancement and vision tasks.

2.2. Deep learning-based methods

Many deep learning-based methods revealed their dominance in image denoising, Lore et al. (2017) proposed stacked denoising self-coding to achieve low-light image enhancement and denoising functions based on denoising self-coding, but only for single-channel grayscale maps. For the multifunctional needs of a single branch or simple neural network that cannot simultaneously perform brightness and contrast enhancement, artifact removal, and noise reduction, Lv et al. (2018) proposed a multi-branch low-light image enhancement network model, MBLLEN, which extracted image-rich features to different levels through a convolutional neural network (CNN), used multiple subnets for simultaneous enhancement, and finally fused the multi-branch output results into a final enhanced image to achieve the multifaceted image quality enhancement, but it lacked a clear theoretical explanation and required a very rigorous computational environment. The DSRL network (Ignatov et al., 2017) proposed by Ignatov et al. combines a composite perceptual error function of color, texture, content, and total change loss to improve performance. In Lee et al. (2018), proposed a deep neural network model capable of reconstructing HDR images from a single low dynamic range image with a progressively enhanced prediction for images at different exposures, but it was difficult to achieve noise reduction while enhancing brightness. Lu

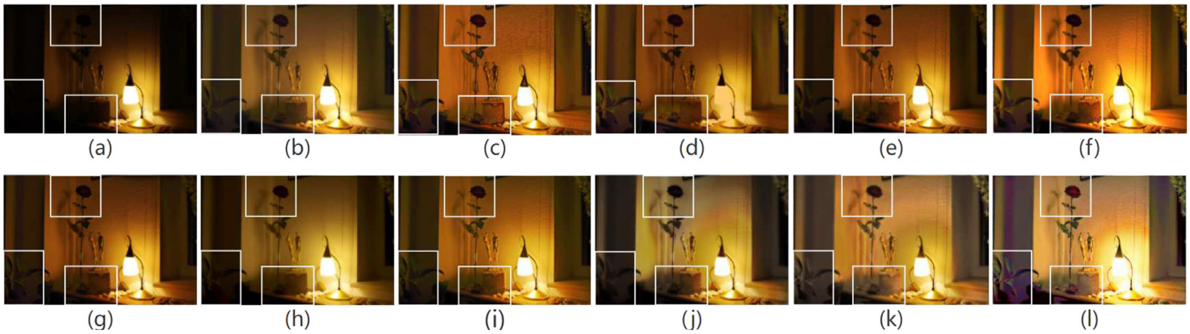

Fig. 2. Results of some existing low-light enhancement methods for "Lamp" in the MEF dataset (Chollet, 2017), (a) input, (b) BIMEF (Cai et al., 2017), (c) Dong (Xuan et al., 2011), (d) NPE (Wang et al., 2013), (e) MF (Chen et al., 2018), (f) LIME (Guo et al., 2017), (g) SRIE (Fu et al., 2016), (h) MBLLEN (Lv et al., 2018), (i) zero-DCE (Guo et al., 2020), (j) kindle (Zhang et al., 2019), (k) KinD+ (Zhang et al., 2021a), (l) Ours. Almost all previous methods have a problem, that is, the ability to enhance very dark areas is very poor, because information is almost lost. But by improving the module for extracting image illumination map, the model can obtain better detail extraction ability, which makes our method do better in enhancing dark areas.

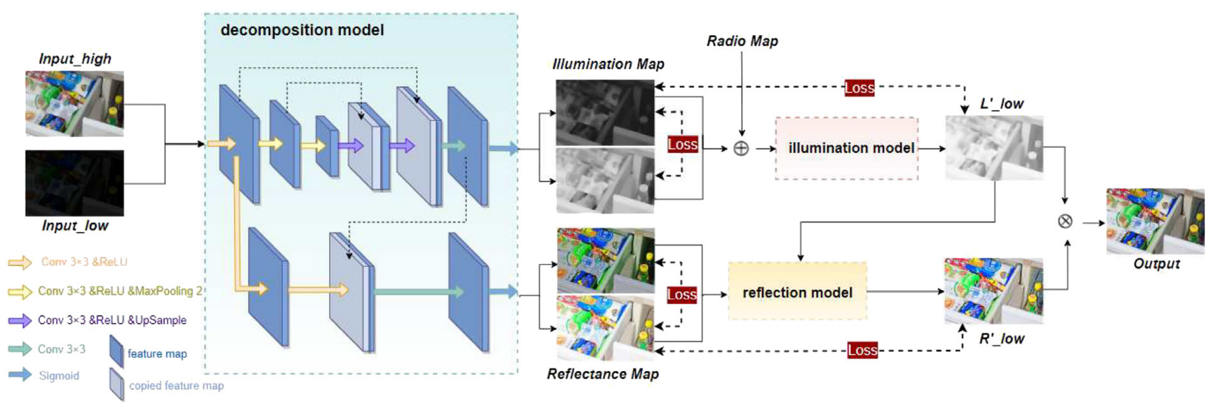

Fig. 3. The network architecture. It can be divided into three modules, which are decomposition module, illumination module and reflection module.

and Zhang (2021) proposed a multi-exposure fusion network called TBEFN for low-light image enhancement, which estimated one transfer function for two branches that could yield two enhancement results. To resolve the difficulty of traditional CNN-based schemes, where elaborately-selected datasets must be first prepared, Jiang et al. (2021) proposed a GAN-based enhancement strategy that could be trained on unpaired datasets, but they sometimes yielded unexpected colors that may look unnatural or over-saturated. In addition, a direct image-to-image transformation is relatively more difficult than that combined with physical priors, this turns out to be more challenging when the noise under low-light conditions is taken into consideration.

There are also many denoising methods (Cho and Kang, 2019; Jin et al., 2020) and unsupervised learning deep learning methods (Ni et al., 2020; Li et al., 2022) for image enhancement, among which a typical one is zero-DCE (Guo et al., 2020), which implemented reference-free training and converted the low-light enhancement problem into a curve transformation problem but did not take into account the noise factor.

3. Methodology

With the rapid development of a computer's ability to process data and image processing algorithms, image enhancement techniques have been studied more. Low-light image enhancement has attracted the interest of many researchers because of its importance in new and advanced applications such as nighttime, unmanned technology, and surveillance.

An ideal low-light image enhancement algorithm should be able to improve the light while also removing the noise from the image. To avoid enhancing the noise in the low-light images, we build a deep learning network with branches for processing illumination and reflection maps, which can also be functionally separated into three modules, namely the decomposition module, the illumination module, and the reflection module. Our overall network structure is shown in Fig. 3, we will go through the pertinent materials one by one in detail in the following parts.

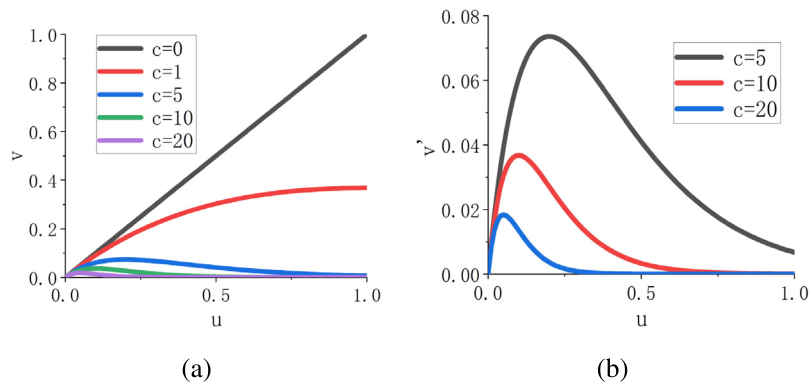

Fig. 4. The behavior of function $v = u \cdot \exp(-c \cdot u)$, on the right hand side is a zoomed-in function plot on the left hand side of where the parameter c controls the shape of this function.

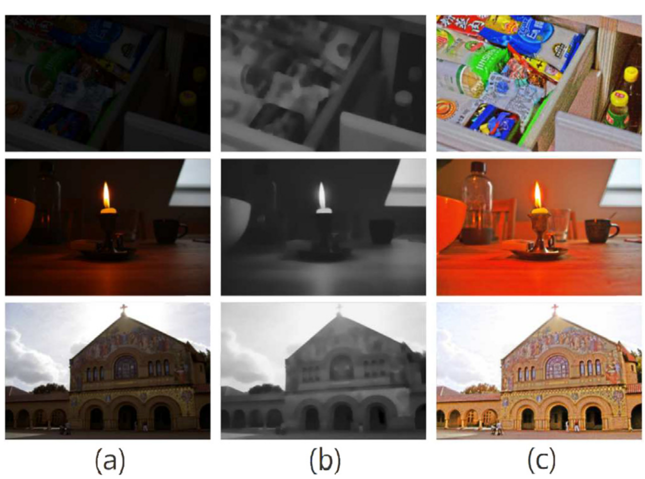

Fig. 5. The effect of decomposition. (a) The low-light image. (b) and (c) represent the decomposition network output, which corresponds to the illumination and reflection maps, respectively.

3.1. Consideration and motivation

The blindness of the illumination map, as stated in the introduction to Section 1, is the reason why many basic image enhancement algorithms fail. As a result, the key to picture improvement is acquiring good and realistic illumination information, i.e., it is critical to properly extract the illumination map from the input low-light image. In Retinex theory, the image can be seen as a composition of two components:

$$S(x, y) = R(x, y) * L(x, y) \tag{1}$$

Where $S(x, y)$ represents the low-light image, $R(x, y)$ is the reflection map describing the intrinsic properties of the observed image, which can be considered a constant independent of illumination, and $L(x, y)$ represents the illumination map indicating the different illumination levels of the observed image. This method of image decomposition can effectively isolate the degraded low-light image space into two smaller subspaces for easier learning. As a result, the Retinex theory-based deconstruction module is critical for the objective job.

Because there is no ground-truth illumination and reflection information to serve as a reference, the conclusion of the decomposition issue is unclear, so it is critical to include prior regularization parameters in the decomposition module. If there is no image degradation, then the reflection maps of images obtained from the same scene should be the same, but the illumination map should be extremely different under different lighting situations, but their structures should be consistent and reasonably simple, according to the Retinex theory. In practice, the degradation that occurs in low-light images should be more severe than the degradation that occurs in normal-light images, and the degradation will be shown in the reflection map. To summarize, while separating the reflection map of a low-light image, we can utilize the reflection map of a normal-light image as a reference for learning.

The contamination of the low-light images, as one of the decomposition module's outputs, is more severe than that of the same region of the normal-light image. According to the Retinex theory, the deteriorated low-light image can be described mathematically as $Input = R \otimes L + P$ where P represents the noise pollution component of the image, which can be obtained by substituting a simple formula:

$$Input = R \otimes L + P = \hat{R} \otimes L = (R + \hat{P}) \otimes L = R \otimes L + \hat{P} \otimes L \tag{2}$$

Where hat R represents the contaminated by noise reflection map and $\hat{P}$ indicates the degradation after image decoupling, with $P = \hat{P} \otimes L$. Assuming that the degradation is caused by additive Gaussian white noise at this point (AWGN), and that $P \sim N(0, \sigma^2)$, $\hat{P}$ i.e., is closely related to the values of L, $\frac{\sigma^2}{Li}$, etc. at each pixel point i. This implies that the recovery of the reflection map cannot be done uniformly across the entire image but must rely on the illumination map for guidance and reference.

3.2. Network design

3.2.1. Layer decomposition net

Decomposing an image into two components is more complex and uncertain, so designing a loss function with suitable restrictions is critical. The easiest part is that we use pairs of normal and low-light images $[I_l, I_h]$ as references. We presume that the reflection maps of multiple lighted images in the same scene are constant, implying that the decomposed reflection map pair $[R_l, R_h]$ should be the same.

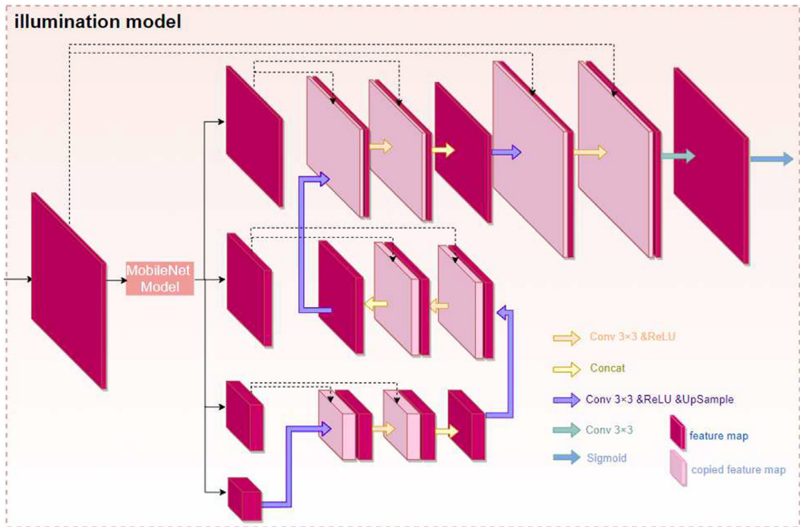

Fig. 6. Illumination module network structure.

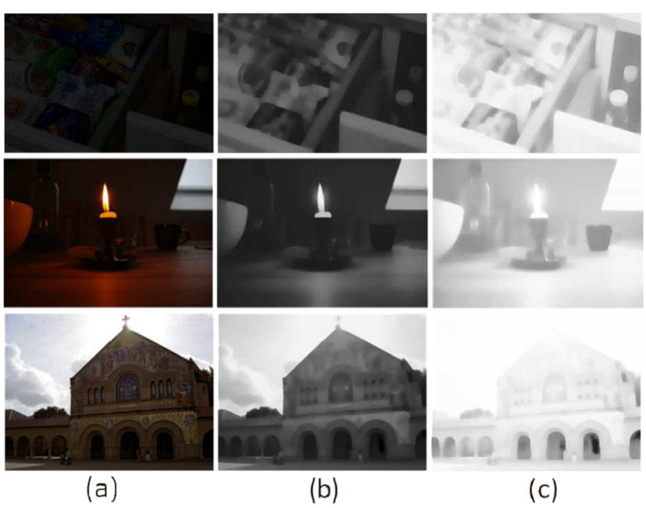

Fig. 7. The effect of the illumination module. (a) Low-light image. (b) The decomposed illumination map. (c) The enhanced illumination map.

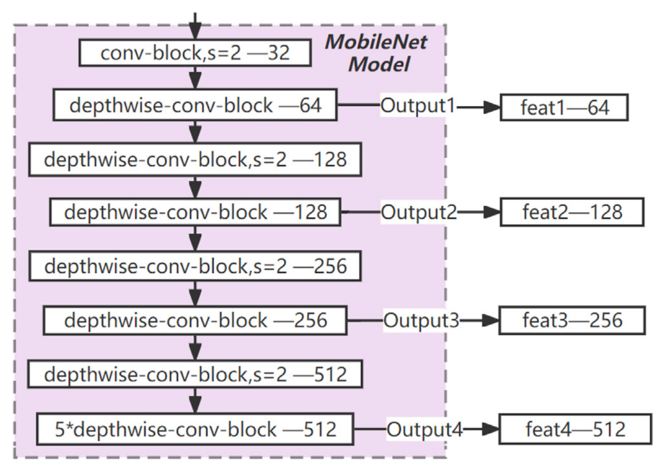

Fig. 8. Mobile-Net network structure.

Furthermore, the decomposed illumination map pair $[L_l, L_h]$ should be segmentally smooth and consistent with one another.

We simply use $L_{rs}^D = \|R_l - R_h\|_1$ to normalize the reflection graph similarity, where $\|\cdot\|_1$ denotes l_1 regularization. The illumination map smoothness is constrained $L_{is}^D = \left\|\frac{\nabla L_l}{\max(|\nabla I_l|,\varepsilon)}\right\|_1 + \left\|\frac{\nabla L_h}{\max(|\nabla I_h|,\varepsilon)}\right\|_1$, in which ∇ denotes the first order derivative containing ∇_x (horizontal) and ∇_y (vertical). To avoid using 0 as a divisor, ε signifies a very small positive constant, and $|\cdot|$ signifies taking the absolute value. This smoothness of the measured illumination is based on the structure of the input image, where the penalty is small at the edge locations and increases at the flat region locations. Because it will use the original input image as a reference and the distribution of light over the smoothed objects should be smooth, this structure is employed to reduce the risk of over-smoothing the structure borders. For mutual consistency loss, when the illumination gradient of the image is small or large, it means that the illumination map is on the smooth object surface (with a more uniform distribution) or at the edge (with a larger distribution difference). We have used $L_{mc}^D = \|M \otimes \exp(-c \cdot M)\|_1$, where $M = |\nabla L_l| + |\nabla L_h|$. That is, the illumination gradient of low light and normal light images and the penalty is large when the illumination gradient is not too small, that is, when the gradient of the two is one big and one small, it indicates that the two illumination maps are different.

Fig. 4 is a graph of the shape of the function of $L_{mc}^D = \|M \otimes \exp(-c \cdot M)\|_1$, where the parameter c controls the function

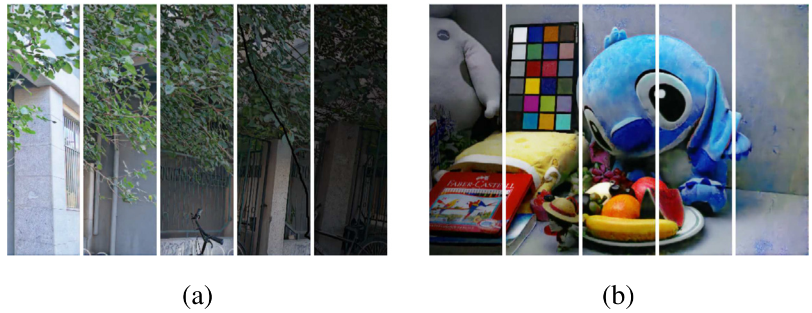

Fig. 9. Manipulation with different light levels by setting r = 5.0,4.0,3.0,2.0,1.0 and r = 0.5,1.5,2.5,3.5,4.5 separately.

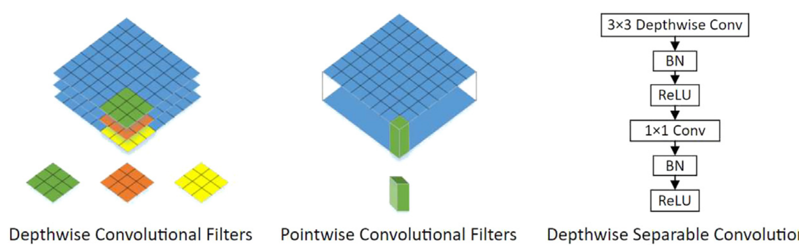

Fig. 10. Depthwise Convolutional Filters, Pointwise Convolutional Filters and Depthwise Separable Convolution Layer Structure.

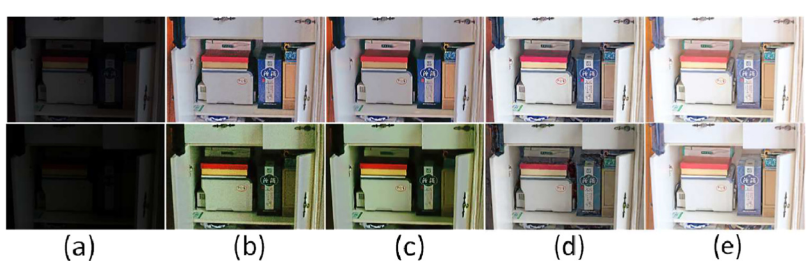

Fig. 11. Comparison of the degree of pollution of the reflection map under different lighting conditions. (a) Low-light image. (b) The decomposed reflection map. (c) The result after BM3D noise reduction. (d) The reflection map output after our reflection module. (e) The reflection map under bright light(the learning objectives of our reflection module).

shape. The penalty increases first and then decreases with the increase of the independent variable until it reaches 0. The shape shows that when $c = 10$, it peaks at $u = 0.1$, indicating that both are relatively weak and have some differences, and should be punished the most severely. As u increases, the penalty gradually decreases, corresponding to the figure, until the penalty is reduced to 0, and our purpose is satisfied, so $c = 10$ is chosen in our code.

In addition, following decomposition, we utilize the reconstruction error function to limit the two images, which implies that the reconstructed image should be as close to the original graph as feasible, so we set the reconstruction error function to:

$$L_{re}^{D} = \left\| I_l - R_l \otimes L_l \right\|_1 + \left\| I_h - R_h \otimes L_h \right\|_1 \tag{3}$$

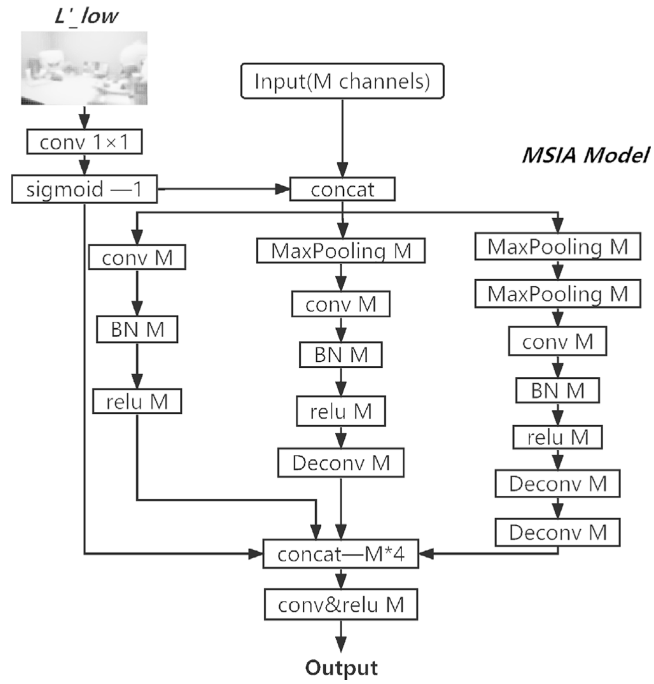

Fig. 12. MSIA Module Structure.

In conclusion, the total loss function of the decomposition module is expressed as:

$$L^D = L^D_{re} + \omega_{rs} L^D_{rs} + \omega_{mc} L^D_{mc} + \omega_{is} L^D_{is} \tag{4}$$

The network structure of the decomposition module is divided into two branches for extracting the reflection map and the illumination map. The reflection branch is composed of a classical 5-layer U-Net structure (Ronneberger et al., 2015), followed by a convolutional layer and a Sigmoid activation function layer (Sun et al., 2021). The illumination branch is composed of two conv+ReLU layers (Aldahoul et al., 2021) and a conv layer on the reflection branch, with a sigmoid layer added at the end. The specific structure is shown in Fig. 3. The effect of the decomposition module is shown in Fig. 5.

3.2.2. Illumination adjustment net

We build a flexible module that can convert low illumination brightness into high illumination brightness. The network structure of our illumination module is shown in Fig. 6, and the specific effect is shown in Fig. 7. Because the amount of light required by each individual varies, we use the ratio parameter r to arbitrarily modify the lighting intensity to fit our demands. By training a pair of illumination maps, although there is no way to know the specific numerical relationship between them, the intensity ratio (L_t/L_s) between their pairs of illumination maps is roughly calculated by setting r. When the illumination is adjusted from low to high, r is greater than one, and when it is adjusted from low to high, r is smaller than one. During the testing phase, the user can adjust the value of r to suit his needs, and the network will first extend r into a single-channel feature map of the same size as L_s, with all elements of the feature map being r. This single-channel feature map is then connected to L_s, which serves as the illumination module's input. This network structure employs the standard feature extraction network structure of Mobile-Net (Xu et al., 2021), and the network layer structure is shown specifically in Fig. 8. We usually take r as a number between 0 and 5. Fig. 9 depicts the effect.

The Mobile-Net model (Shen et al., 2018) is a lightweight deep neural network proposed by Google for embedded devices like cell phones, based on the Depthwise separable convolution (Chollet, 2017) principle. In the Mobile-Net structure, the Depthwise-conv block is a layered convolution, followed by a batch normalization layer (Liang et al., 2021) and a ReLU layer, and then a 1×1 convolution for channel processing, as shown in Fig. 10.

The loss function of the illumination module is $L^A = MSE(\hat{L}, L_k) + MSE(\nabla\hat{L}, \nabla L_k)$, where L_k is either L_l or L_h and $\hat{L}$ is the adjusted illumination map output by the illumination module. Where ∇ denotes the first-order derivative containing ∇x(horizontal) and ∇y (vertical), MSE is the mean square error value.

3.2.3. Reflectance restoration net

In general, the degradation in the reflection map decomposed from the low-light image is more obvious than in the reflection map decomposed from the normal image, in Fig. 11, the light conditions in the lower row are darker than in the upper row. What can be seen is that the reflection map in lower light corresponds to a heavier degradation level.

The degradation usually appears unevenly in the reflection map, which is closely related to the illumination map, so we use a multi-scale illumination attention(MSIA) module in the reflection module to deal with the degradation appearing in the decomposed reflection map. The MSIA structure is shown in Fig. 12, which consists of the illumination attention module layer and the multi-scale module layer.

The whole network of the reflection module is composed of 10 convolutional layers and 4 MSIA module layers. The structure is shown in Fig. 13.

The loss function of the reflection module used here is $L^R = L^R_{mse} + L^R_{dsim}$ where The first term L^R_{mse} represents the mean square error of R_h and $\hat{R}$, and $\hat{R}$ represents the reflection map after reconstruction, the second term L^R_{dsim}=$1 - SSIM(R_h, \hat{R})$ is used to evaluate the similarity of R_h and $\hat{R}$ based on the structural similarity SSIM (Sabilla et al., 2021). The effect is shown in Fig. 14. The reflection module can well retain the details of the reflection map and has a certain noise reduction effect, such as the first drawer edge texture. The comparison of the noise reduction effect will be introduced in detail in the experimental section.

4. Experimental validation

In this section, we detail the experimental details to demonstrate the validity of the method. Firstly, we present the data set and experimental details as well as the evaluation metrics. Secondly, we tested the performance of the network modules to prove the effectiveness of each module. Finally, we perform a qualitative and quantitative comparison with other state-of-the-art low-light enhancement methods to compare the performance of our method.

4.1. Network training strategy

4.1.1. Dataset and implementation details

Low-light image enhancement has been a relatively popular research direction, but there are fewer paired-based public datasets in real-world scenarios. We use a publicly available low-light dataset, the LOLdataset (Wei et al., 2018) with a total of 500 paired images for network training. In the validation process, we randomly selected 50 paired images from the LOLdataset for testing. In addition, we also used the benchmarks LIME (Guo et al., 2017), NPE (Wang et al., 2013), MEF (Chollet, 2017), DICM (Fu et al., 2016) for testing.

The experiment uses Python to write algorithm code, uses Adam (Camacho et al., 2020) as the optimizer, is implemented based on the Tensorflow (Ertam and Aydin, 2017) framework, and trains under NVIDIA GeForce RTX 3090 GPU. The batch size of the three networks is set to 10, and the initial learning rate is set to 1e–4. The three modules are trained separately, and the number of training rounds is set to 2500, 2000, and 1000, respectively. Compared to the end-to-end method, using module training separately not only shortens the training process but also facilitates debugging if there are errors in the training. If a module outputs errors, we can artificially correct the errors by observing the output of each module.

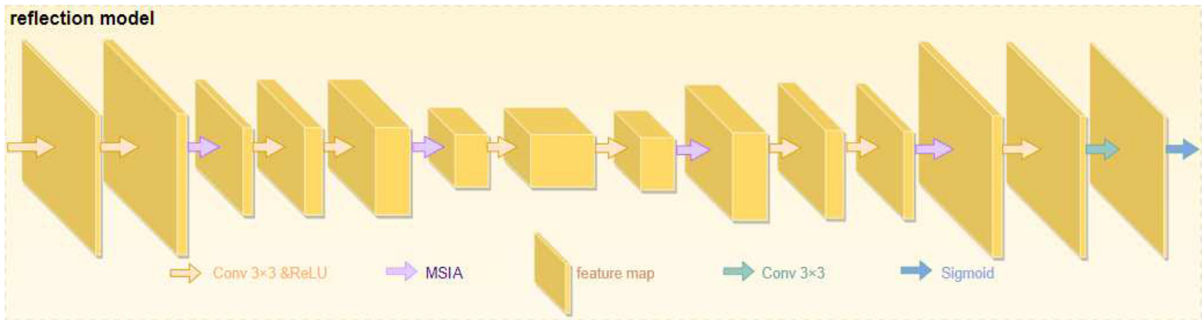

Fig. 13. Reflection module network structure.

Table 1
Comparison of the real low-light dataset in Fig. 19 with the state-of-the-art low-light enhancement methods in metrics NIQE, NLIEE.

Metrics	BIMEF	Dong	NPE	MF	LIME	Retinex-Net	DLN	Zero-DCE	Kindle	KinD+	ours
NIQE↓	2.74	2.90	**2.64**	2.88	3.03	5.00	3.30	3.14	3.52	3.20	*2.73*
NLIEE↑	49.39	46.20	49.97	50.79	55.59	50.16	48.25	50.09	50.23	*54.26*	**57.53**

Table 2
Quantitative analysis of the LOLdataset in terms of PSNR, SSIM, LIEQA, NIQE and EI.

Metrics	PSNR↑	SSIM↑	LIEQA↑	NIQE↓	EI↑
BIMEF	13.54	0.56	0.21	7.75	6.37
Dong	15.17	0.44	0.08	7.75	6.94
NPE	15.96	0.44	0.09	8.63	7.09
MF	15.87	0.47	0.11	9.09	6.96
LIME	15.81	0.41	0.10	8.67	*7.23*
Retinex-Net	15.49	0.39	0.06	9.28	6.86
DLN	18.44	0.68	0.20	6.41	6.19
Zero-DCE	16.54	0.47	0.13	8.05	7.00
Kindle	19.80	0.77	*0.37*	*4.70*	6.90
KinD+	*20.01*	**0.82**	0.33	4.71	6.68
ours	**20.93**	*0.78*	**0.37**	**4.25**	**7.30**

Table 3
Quantitative analysis of each data set in terms of AG.

Datasets	LOL	DICM	LIME	MEF	NPE
BIMEF	6.80	7.60	8.89	7.13	8.42
Dong	12.11	*10.65*	*12.73*	**11.32**	*12.52*
NPE	**15.04**	8.79	10.31	9.30	8.67
MF	*12.73*	8.82	10.63	8.93	10.02
LIME	5.08	5.80	7.14	4.83	5.92
Retinex-Net	5.32	7.31	10.19	7.71	9.27
DLN	8.64	6.69	8.59	6.82	7.55
Zero-DCE	7.73	8.09	10.07	8.20	9.54
kindle	5.20	6.42	7.66	6.38	8.44
KinD+	6.21	7.99	11.32	8.51	10.89
ours	8.66	**11.06**	**14.24**	*10.93*	**13.11**

4.1.2. Evaluation criteria

Perceptual image quality assessment aims to quantify the human perception of image quality (Zhai and Min, 2020; Min et al., 2021). In recent years, a large number of studies have developed in this direction. Min et al. (2018a) proposed, by deriving a pseudo-reference image (PRI) image, a blind PRI-based (BPRI) evaluation framework. Min et al. (2018b) generated a series of PRI by further degrading the distorted

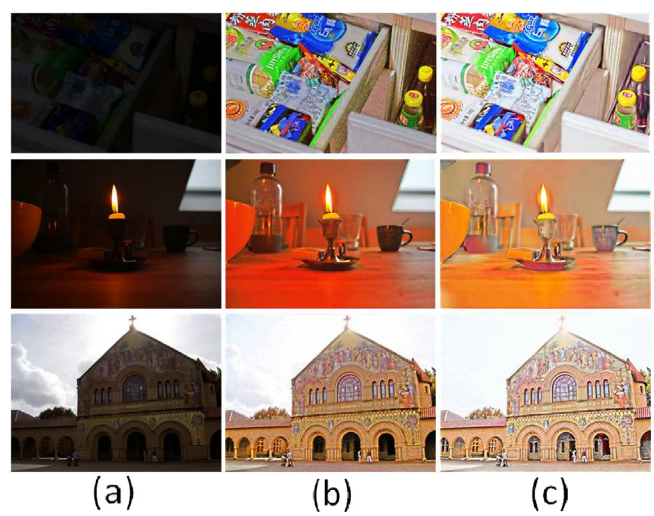

Fig. 14. The effect of the reflection module. (a) Input. (b) Reflection map by decomposition. (c) Reflection map after the reflection module.

image and then used local binary patterns to measure the similarity between them to evaluate their quality. Min et al. (2020) conducted an in-depth exploration of the problem of assessing the quality of audio and video signals. In addition, some image quality evaluation indicators are related to the dehazing quality evaluation. Min et al. (2018c) evaluated image dehazing algorithms by assessing the quality of real dehazed images directly. The low-light image enhancement effect evaluation is similar to the dehazing quality evaluation. For low-light image enhancement evaluation, either a synthetic low-light and normal-light image pair or a realistic low-light image could be

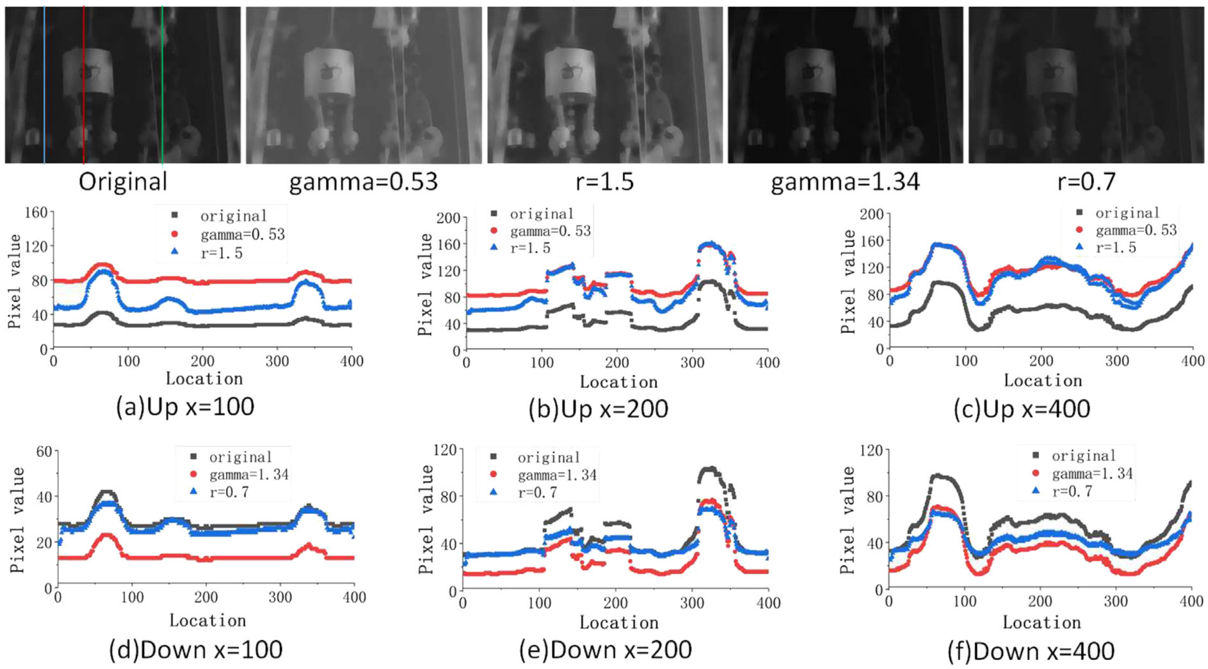

Fig. 15. The effect of the enhanced illumination map is compared with gamma correction. We selected an image from the LOL dataset and compared their pixel value sizes at $x = 100$, 200, and 400 of the image, respectively.

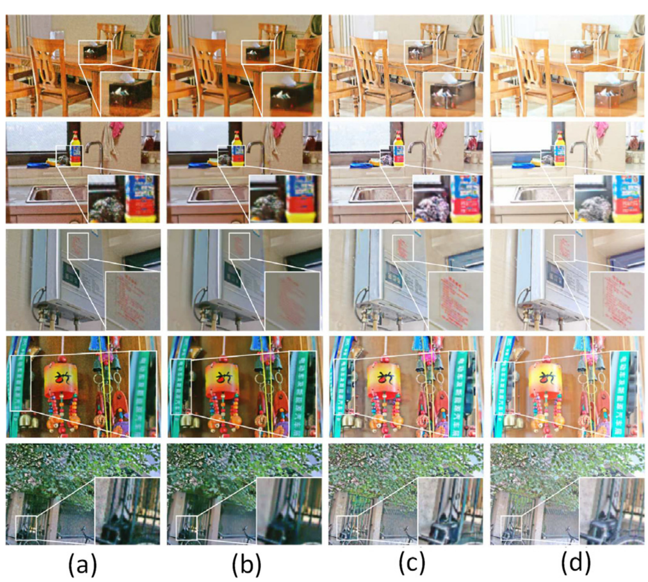

Fig. 16. The effect of the reflection module. (a) Reflection map of the low-light image decomposition, (b) Reflection map after passing the reflection map through BM3D noise reduction. (c) Reflection map after passing the reflection map through the reflection module. (d) Reflection map of the normal light decomposition(the learning object of the reflection module).

used. A synthetic hazy/normal image pair or a realistic hazy image could be used for dehazing. Min et al. (2019) evaluated image dehazing algorithms using synthetic hazy images and proposed a full-reference dehazing algorithm (FR DHA) evaluation method considering image structure recovery, color rendition, and over-enhancement of low-contrast areas. To evaluate the enhancement results more objectively, we used the image quality evaluation metrics for low-light image enhancement, divided into the non-reference evaluation and reference evaluation measures, to evaluate the quality of low-light image enhancement more comprehensively.

The reference-based evaluation metrics require that the enhanced low-light image be compared with the reference image. We use the peak signal-to-noise ratio PSNR, structural similarity SSIM (Sabilla et al., 2021) and low-light image enhancement quality assessment LIEQA (Zhai et al., 2021) to measure the gap between the enhanced image and the ground truth image. PSNR is used to evaluate image fidelity by calculating the mean square error between the enhanced low-light image and the ground truth image. The higher the PSNR, the smaller the image distortion. SSIM considers the images' similarity in terms of brightness, contrast, and structure, respectively. The larger the value of SSIM, the more similar the structure of the enhanced image is to the ground truth image, and the smaller the loss. LIEQA is a new FR low-light image enhancement quality assessment index that evaluates image quality from four aspects: luminance enhancement, color rendition, noise evaluation, and structure preservation. The higher the LIEQA, the better the image quality. Based on the evaluation metrics of no-reference, we used the Natural Image Quality Evaluator (NIQE) (Mittal et al., 2013), Comprehensive image quality index NLIEE (Zhang et al., 2021b), the Average Gradient (AG) and the image information entropy (IE) to evaluate the quality of no-reference images. NIQE is a distance metric between the computed model statistics and the enhanced low-light image based on the spatial domain natural scene statistics model. When the NIQE value is lower, it means that the closer to the natural image, the higher the image quality. The NLIEE is mainly assessed from four key aspects: light enhancement, color comparison, noise measurement, and structure evaluation. The larger the value, the better the comprehensive image quality. The Average Gradient (AG) is used to measure the clarity of the fused image, and the larger its value, the higher the image clarity and the better the fusion quality. The image information entropy (IE) reflects the richness of the image information, and the larger its value, the richer the image information and the better the quality.

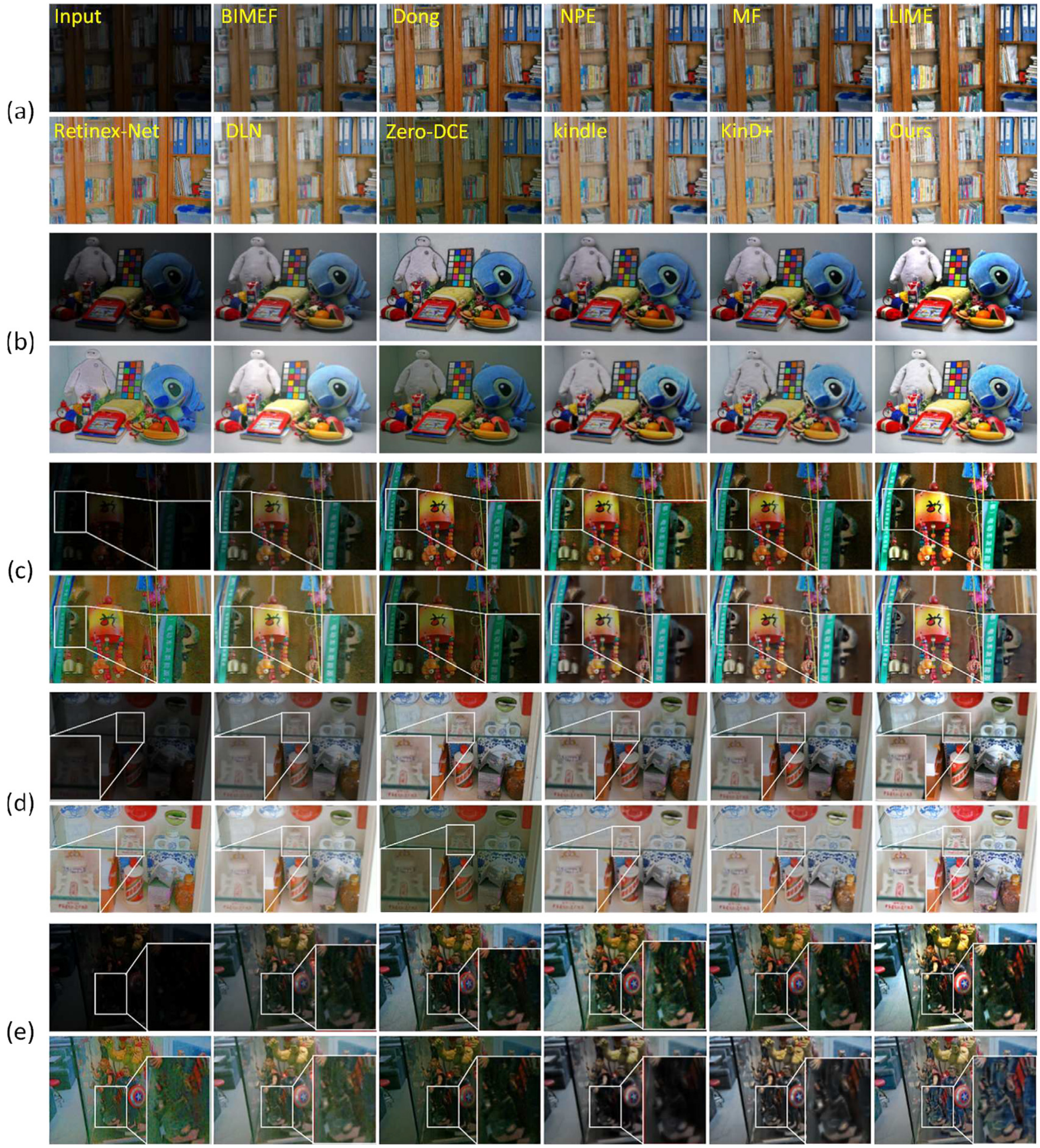

Fig. 17. Visual comparison of the enhanced effect of 5 random images in the LOL dataset. The algorithms for the comparison are BIMEF (Cai et al., 2017), Dong (Xuan et al., 2011), NPE (Wang et al., 2013), MF (Chen et al., 2018), LIME (Guo et al., 2017), Retinex-Net (Wei et al., 2018), DLN (Wang et al., 2020b), zero-DCE (Guo et al., 2020), kindle (Zhang et al., 2019), KinD+ (Zhang et al., 2021a) and ours.

4.2. Network performance testing

4.2.1. Illumination adjustment net

Fig. 15 shows the difference between the light adjustment and gamma correction learned by the illumination module. To be fair, we set $gamma = \frac{\|\log(\hat{L})\|_1}{\|\log(L_s)\|_1}$ to make the comparison. We consider two cases of adjustment: brightening and darkening the illumination map. To show the differences more clearly, we plot the pixel values at pixel points $x = 100$, 200, and 400, respectively. Regarding the light-up case, we can observe that in the bright areas, we learn more or have the same intensity enhancement as the gamma-corrected results, while in the dark areas, we will have less intensity enhancement. Regarding the light-down case, the opposite trend occurs. In other words, less light is enhanced in the darker areas, while more or the same light is enhanced in the brighter areas. In general, we learn that in relatively bright areas, intensity increases more, or about the same as gamma correction, while in dark areas, intensity increases less. And the brightness of the image after gamma correction, whether in dark areas or bright areas, will not be adjusted according to the brightness of the area. The overall increase in light is the same. That is, gamma correction is a general enhancement of the whole image, whereas, according to the way we learned, it will

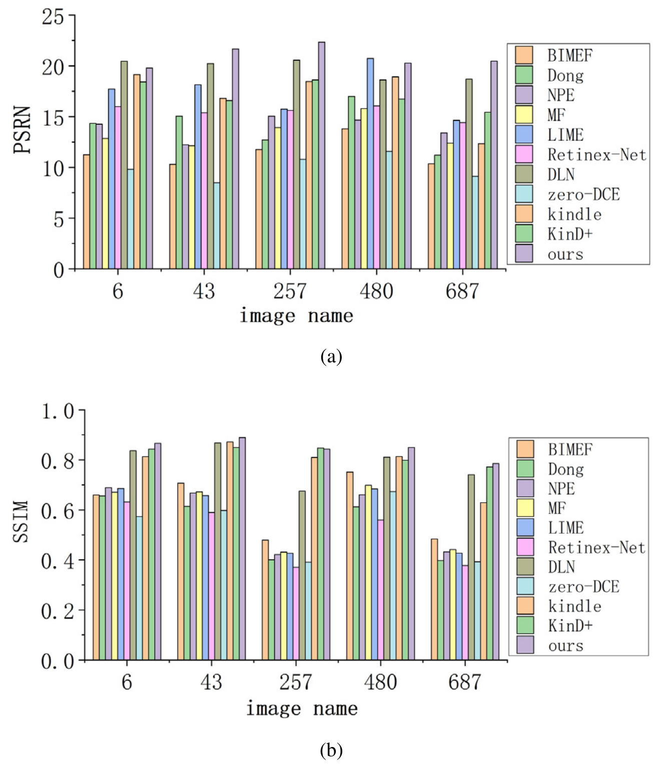

Fig. 18. PSRN and SSIM of 5 random images in the LOLdataset.

be adjusted according to the degree of light and darkness of the image area. Light areas will increase the light, and dark areas will decrease the light. In summary, the way we learned is more realistic.

4.2.2. Reflectance restoration net

Although BM3D can remove the noise of the reflection map well, the blurring appears in the details, but the effect of our reflection module, as shown in Fig. 16, can be seen in the details and texture of the drawer box in the first row of the image, the bottle part of the detergent in the second row, the red lettering part of the water tank in the third row, the white lettering part on the green thick strip in the fourth row, and the front part of the bicycle in the fifth row. Fig. 16(c) has a brightening effect because the corresponding illumination map is fused in the reflection module. Our reflection module retains the details and brightness well. So the reflection module's visual de-noising result is superior to that of BM3D and is closer to the reflection map of the normal-light images.

4.3. Performance evaluation

In this section, we will qualitatively and quantitatively go through our network and compare it with ten state-of-the-art methods, all of which are BIMEF (Cai et al., 2017), Dong (Xuan et al., 2011), NPE (Wang et al., 2013), MF (Chen et al., 2018), LIME (Guo et al., 2017), Retinex-Net (Wei et al., 2018), DLN (Wang et al., 2020b), zero-DCE (Guo et al., 2020), kindle (Zhang et al., 2019), and KinD+ (Zhang et al., 2021a).

4.3.1. Comparison on a single image

We can observe in Fig. 17 that our method outperforms all the other methods in terms of brightness, even though the two methods, LIME (Guo et al., 2017) and DLN (Wang et al., 2020b), look as if they are competing, for LIME suffers from over-enhancement and some degree of color distortion, and DLN suffers from unclear or even missing details and textures. The results of the KinD+ and our method are very similar, but our results enhance the brightness of dark areas in the images with the assurance that textures are not missing and colors are not distorted. Degradation is also better handled. For example, in Figs. 17(c), 17(d), and 17(e), the parts that are enlarged by the white box, our method preserves the best details without distorting the colors, and the enhanced brightness are excellent. In Fig. 17(c), we can see the white letters on the green, the thick strip on the left, and the dark area behind the wind chimes. Our method ensures both color saturation and contrast with details, while the methods of LIME (Guo et al., 2017), Retinex-Net (Wei et al., 2018), kindle (Zhang et al., 2019), and KinD+ (Zhang et al., 2021a) seem to lose some of the texture of the wall in the shadow part of the wind chimes, and NPE (Wang et al., 2013) and DLN (Wang et al., 2020b), although the texture detail of the wall is maintained, the color distortion appears in the part of the white letters on the left side. This phenomenon is more evident in the dark parts of the legs of the hand puppet in Fig. 17.(e). To better see the effect of the five images, we show their PSRN and SSIM values in histograms for better comparison and observation. As shown in Fig. 18, our values are all superior to the other methods.

We randomly extracted five images from four non-reference datasets, DICM, LIME, MEF, and NPE, and tested them using NIQE and NLIEE non-reference quality evaluation indicators, respectively, to compare the effects of our method and other low light level enhancement methods on non-reference data sets. As shown in Fig. 19, it can be observed from the enlarged detail that our method is very effective in enhancing the dark areas in the low-light image. Compared with other enhancement algorithms, our method can maintain good color saturation and detail while enhancing the image's brightness and can also effectively suppress the amplified noise. Although the Retinex-Net method can also effectively enhance image brightness, it is accompanied by severe color distortion and amplified noise. KinD+ seems very competitive, but there is a degradation phenomenon. From a single rendering, we can see that our method has great advantages over the other ten algorithms. The image quality recovered by various enhancement methods is evaluated by the non-reference image quality evaluation indicators NIQE and NLIEE (Zhang et al., 2021b). The best results in Table 1 are shown in bold, and the suboptimal results are in italics. The statistical results show that our method is superior to most outstanding low light enhancement methods in these two evaluation indicators, especially with the highest NLIEE, which proves that our method has good performance in non-reference datasets.

4.3.2. Comparison on dataset images

We measured our method using several widely adopted datasets, including LOLdataset (Wei et al., 2018), LIME (Guo et al., 2017), NPE (Wang et al., 2013), MEF (Chollet, 2017), DICM (Fu et al., 2016), and six metrics for quantitative comparison, including PSNR, SSIM, NIQE (Mittal et al., 2013), NLIEE (Zhang et al., 2021b) and IE, AG (Wang et al., 2020a). The latter four do not need corresponding images for reference, and to measure the average gradient AG, we used different datasets without paired reference images.

Table 2 shows the results of each metric for all the excellent low-light enhancement algorithms listed on LOLdataset. The best results are shown in bold font, and the second-best results are shown in italic font. Each low-light image used for testing has a normal image as a reference, so both PSRN and SSIM values can be used for comparison. Numerically, our network results are significantly better than the other methods in most metrics; the LIEQA of our method ranks first among the eleven low light enhancement algorithms compared, which proves that our method has obvious advantages in image contrast, structure, and noise processing, and our method is in the top three for both the reference metrics PSNR and SSIM, showing a great advantage. And it also performs well with the non-reference metrics NIQE and EI.

For the LIME (Guo et al., 2017), NPE (Wang et al., 2013), MEF (Chollet, 2017), and DICM (Fu et al., 2016), since they are all datasets without paired reference images, we only compare the AG values of

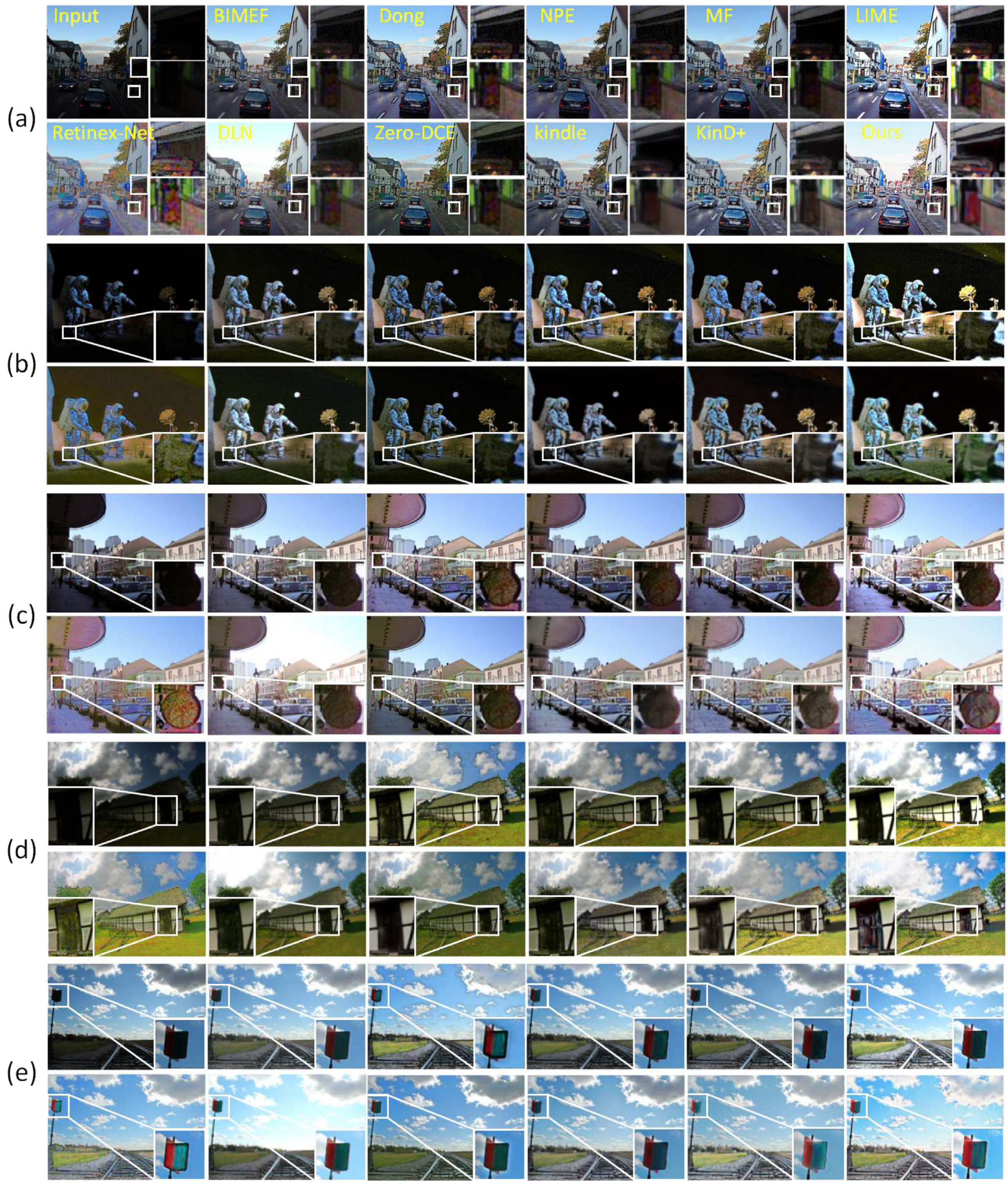

Fig. 19. Comparison of the visual effects of our method and other low-light enhancement methods in the other datasets. (a) from LIME, (b) and (c) from DICM, (d) from MEF, (e) from NPE.

each algorithm, which are BIMEF (Cai et al., 2017), Dong (Xuan et al., 2011), NPE (Wang et al., 2013), SRIE (Fu et al., 2016), MF (Chen et al., 2018), MBLLEN (Lv et al., 2018), TBEFN (Lu and Zhang, 2021), DLN (Wang et al., 2020b), Zero-DCE (Guo et al., 2020), kindle (Zhang et al., 2019), and KinD+ (Zhang et al., 2021a), as shown in Table 3, the best results are shown in bold font and the second best results are shown in italic font. To be able to better visualize the enhanced results, we plotted the AG values in the table as a dotted line plot, as in Fig. 20. Although our method does not show good advantages in the LOLdataset, it still shows obvious advantages for other datasets.

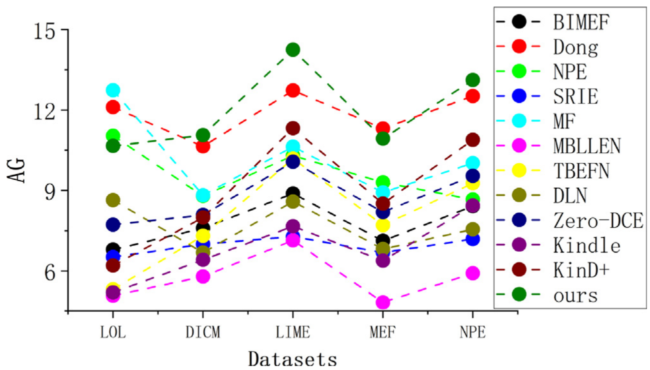

Fig. 20. Dotted line plots of AG values plotted on each data set.

5. Ablation study

The performance of the algorithm depends on the network structure and the setting of the loss function, and this section evaluates the structure of the illumination module and the effectiveness of the loss function.

5.1. On the illumination net network architecture

With the constant loss function, we separately tested the network structure shown below as the structure of the illumination module. Whereas Fig. 21(a) is composed of three conv+ReLU layers and three resizing modules that resize the image and then perform concat and a conv layer. Fig. 21(b) is composed of three conv+ReLU layers and one conv layer, a Sigmoid layer. Fig. 21(c) adopts the same structure depth as our method, but the convolution layer uses the basic conv layer.

For a fair comparison, we used the best possible parameters for each candidate network. Since there is no reference for quantitative measurement of performance, we show the visualization results for different network architectures in Fig. 22 and ensure that the illumination map of the input low-light image is uniform, i.e., in (a) of the figure. As can be seen from the figure, the illumination adjustment results for (b) lose a lot of details and show problems such as overexposure and lack of smoothness, compared with the better illumination adjustment results for (c) and (d). As for (c) and (d), the details of (d) are more obvious in some dark areas, and the light added to the dark areas is less than the light enhanced in the bright areas, such as the stool texture part of the first image, the texture by the window of the second image, the water tank display area of the third image, the left wind chime area of the fourth image, and the leaf part of the fifth image. The results show our enhanced effects are clearer, more detailed, more textured, and more closely match the illumination map of the normal light image decomposition, thus confirming the better learning of our network.

The dark areas of KinD+ cannot be well enhanced, which leads to the loss of some details. After a lot of thought and experiments, we proposed to improve the network structure of the illumination module to make the illumination map of the low-light image feature extraction better. But because the network depth is increased and the network structure is complicated, we also refer to the lightweight network Mobile-Net, It is proposed to use the depth separable convolution instead of the ordinary convolution to reduce the network parameters and make the network lightweight. By increasing the ablation experiment and comparing the parameters in Table 4, we can see that the network depth of KinD+ is enhanced to the same depth as our method. The parameter of the illumination module is 472.5K, while the parameter of the illumination module in our network is only 280.2K. The effect is also shown in Fig. 22. The illumination map of our network, which is closest to the illumination map of a normal-light image, further proves the difference and improvement between our network and KinD+.

Table 4
Comparison of parameters used in proposed vs. KinD+*.

Method	Params
KinD+	19.4K
KinD+*	472.5K
Ours	280.2K

5.2. On the illumination net loss function

In Fig. 23, it can be seen that using only (b) of $MSE(\hat{L}, L_k)$, both images are not smooth enough, as shown in the whole image, where the contrast is also excessive at the position of the wind chimes and the left satin ribbon in the first image, and at the position of the edge of the window and the basin in the second image. And the result of using only $MSE(\nabla\hat{L}, \nabla L_k)$ again leads to the effect of over-smoothing and overexposure of the illumination map, such as the pattern on the wind chimes in the first image and the position of the window and the top row of bowls in the second image. And (d), we used the sum of both, and by comparing with (e), we found a significant improvement in the learning effect, which ensures the details while making the images smooth. Therefore, in this paper, we use the sum of the two as the loss function of the illumination module.

6. Conclusions

In this work, we build a depth convolutional neural network for enhancing low-light images. Our key idea is based on Retinex theory, which does not require obtaining real information about image illumination and reflection maps, decomposing a low-light image into illumination and reflection maps, and then processing them for illumination adjustment and reflection reconstruction, respectively, to recover clear details, distinct contrast, and vivid colors, and finally fusing them into a high-quality clear image with rich texture and sharp edges.

Our attention is mainly focused on providing a more efficient illumination module to improve the shortcomings of previous illumination maps that were under-illuminated in certain dark parts and to better adjust of the illumination map. With a clear physical interpretation of our structure, extensive experiments were conducted on low-light image datasets to demonstrate the clear advantages and effectiveness of our method under both subjective and objective metrics by comparing our method visually and quantitatively for five metrics with the state-of-the-art ten methods. A large number of experiments proved that our solution is qualitatively and quantitatively effective, but since the evaluation of low-light image enhancement is very similar to the dehazing quality evaluation, the validation of enhanced images using the dehazing quality assessment needs to be further studied and analyzed, and this model can be considered for other enhancement tasks, such as low-light video enhancement.

CRediT authorship contribution statement

Manli Wang: Conceptualization, Project administration, Writing – review & editing. **Jiayue Li:** Methodology, Software, Validation, Formal analysis, Data curation, Resources, Writing – original draft, Writing – review & editing, Visualization. **Changsen Zhang:** Supervision, Writing – review & editing.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Li Jiayue reports financial support was provided by Henan Polytechnic University.

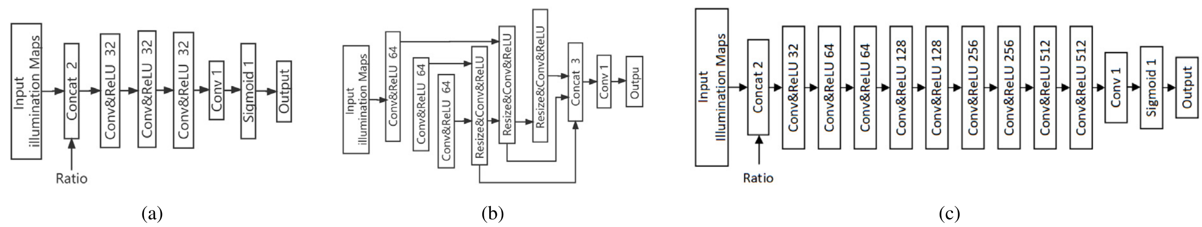

Fig. 21. Different candidate architectures a, b and c.

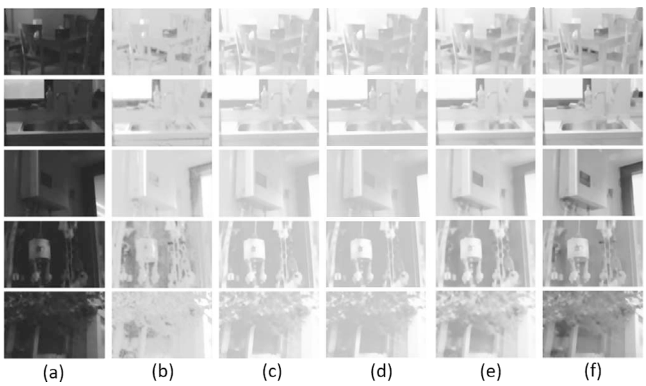

Fig. 22. Visual comparison of light maps of different architectures of five images in the LOLdataset. (a)Illumination map of the input, (b), (c), and (d) correspond to the output illumination maps of the architecture of Fig. 21(a), (b), and (c) as the network structure of the illumination module (e) The illumination map of the illumination module through our method (f) The illumination map under normal light for reference.

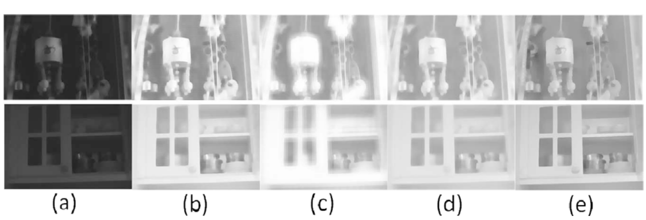

Fig. 23. (a)Illumination map for the output low-light image. (b)Illumination map for only $MSE(\hat{L}, L_k)$ is used as the loss function. (c)Illumination map for only $MSE(\nabla\hat{L}, \nabla L_k)$ is used as the loss function. (d)Illumination map for $MSE(\hat{L}, L_k) + MSE(\nabla\hat{L}, \nabla L_k)$ is used as the loss function. (e)Illumination map for the bright light image.

Data availability

No data was used for the research described in the article.

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